1 Samjha AI – Overall Model Framework

(How everything works together)

User
(Chat / Task / File / Voice)

Samjha AI Interface (Existing Website)
- Chat UI
- Files
- Voice (future)
- Tools
- Projects

Existing website will not be harmed.
- Same UI
- Same login
- Same chat
- New features added as module (SCEM)

User Request (Natural Language)
e.g. "Make a research report"

SCCL (Samjha Cognitive Control Language)
Convert user request into structured task
Define goal, steps, tools, permissions, verification

Natural Language ↕ SCCL Task
e.g. research_brief.sccl

SCEM (Cognitive Execution Model)

APM (Adaptive Planning)
- Break task
- Choose steps
- Handle failures

CPM (Context & Memory)
- Retrieve past chats
- Project knowledge
- Relevant files

LPM (Learning & Performance)
- Evaluate results
- Save feedback
- Improve future plans

VEM (Verification Engine)
- Check accuracy
- Validate sources
- Ensure safety

Memory Store
- Chat history
- User files
- Project data
- Preferences
- Vector DB

Tool Execution
- Web Search
- File Read/Write
- Code Execution
- Image/Video
- Browser (future)
- Other Tools

Model & Tool Runtime (Using Existing Models)
- Open Source Models (e.g. Llama, Mistral)
- Existing Working Model (Current Chat)
- Other Models (Via routing) (Open & Free)

Final Response to User
Text · Document · Code · Image · Plan · Action · Other Result

2 Step-by-Step Working Process

(From User Request to Final Answer)

1 User gives a request

"Research how AI is used in education and create a report with sources."

Input can be text, file, image, voice, or a complex task.

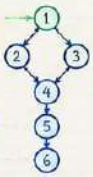
2 Convert to SCCL

SCCL Task (Structured)
```
goal:        "Create research report"
steps:       [search, read, analyze, write]
tools:       [web.search, file.read, document.write]
constraints: "Use reliable sources"
verify:      "Each claim must have source"
```

Natural language is converted into SCCL so the task is clear, safe and executable.

3 APM creates a plan

Execution Plan (DAG)
1. Search for sources
2. Read and extract key points
3. Compare information
4. Write structured report
5. Verify sources
6. Give final output



APM breaks the task into steps and creates a dependency graph.

4 CPM retrieves relevant memory



- Previous chats
- Project files
- User preferences
- Relevant knowledge

CPM finds useful information from your past work and approved memory.

5 Execute each step using tools + models

Search → Read → Analyze → Write

Each step is executed using allowlisted tools and models.

6 VEM verifies the result
- ✅ Are all claims supported by sources?
- ✅ Is the information accurate?
- ✅ Is it within allowed permissions?
- ✅ Any unsafe or harmful content?

VEM checks the output. If something fails, it can re-run steps or ask for your approval.

7 LPM learns and improves



- Stores task result and feedback
- Finds what worked and what failed
- Improves future planning and tool selection

LPM uses feedback to improve future responses without changing the main chat behavior.

8 Final response to user

Report · Sources · Steps · Insights

You get a verified, well-structured response with sources and clear steps.

3 SCCL Language Structure & Example

(How tasks are written and executed)

What is SCCL ?
- ✅ A simple, structured language to define AI tasks.
- ✅ It tells the system: goal, steps, tools, permissions, and verification.
- ✅ It is not a programming language like Python.
- ✅ It is a control language for Samjha AI (SCEM).

SCCL Grammar (Simplified)
```
task <name> {
    goal: <text>
    context: <text>
    step <id> {
        tool: <tool_name>
        input: <expression>
        requires: <dependency>
        expect: <output_type>
    }
    verify <id>: <condition>
    policy <id>: <rule>
}
```

Main Elements
- task : Define a task
- goal : What to achieve
- context : Additional info
- step : A single step
- tool : Which tool to use
- input : What to give as input
- requires : Dependency
- expect : Expected output
- verify : Check result
- policy : Safety / permission rule

Example SCCL Task
```
task ResearchBrief {
    goal: "Create a research brief on AI in education"
    context: "Use reliable and recent sources"

    step search_sources {
        tool: web.search
        input: goal
        requires: none
        expect: source_list
    }
    step analyze_content {
        tool: document.analyze
        input: search_sources.result
        requires: search_sources
        expect: key_points
    }
    step write_report {
        tool: document.compose
        input: analyze_content.result
        requires: analyze_content
        expect: report_document
    }
    verify sources_present: every_claim_has_source(write_report.result)
    verify no_hallucination: check_factual_consistency(write_report.result)
    policy permissions: allow([web.search, document.compose])
}
```

This task will:
1. Search the web
2. Analyze content
3. Write a report
4. Verify sources
5. Follow safety rules

How This Beats New Models ?
- ✅ Better task completion (not just text generation)
- ✅ Structured planning with verification
- ✅ Persistent memory and project context
- ✅ Tool use with permissions and safety
- ✅ Lower cost using open models + smart routing
- ✅ Clear and inspectable execution (SCCL)
- ✅ Easier fine-tuning of planner/workflow later
- ✅ User control and privacy first
- ✅ Real-world workflows (study, research, coding, analysis, automation)

Key Advantage

Not a bigger model, but a smarter system that understands, plans, executes, verifies and improves real tasks reliably.